Asynchronous Two-Tier Spacecraft Anomaly Triage Using Masked Diffusion Under Radio Blackouts

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Abstract*—Detecting anomalies in spacecraft telemetry on onboard edge processors is heavily constrained by Size, Weight, and Power limitations and the physical reality of discontinuous data transmissions. Standard deep learning anomaly detectors rely on continuous sliding-window evaluations, creating severe computational overhead and forcing numerical interpolation across routine communication gaps, such as 149-second radio blackouts, which induces artificial high-frequency transients and false alarms. In this paper, we propose an asynchronous two-tier anomaly triage and root-cause diagnostic cascade tailored for resource-constrained aerospace telemetry. Layer 1 operates as an ultra-fast edge triage reflex, utilizing a per-channel tail-calibrated Isolation Forest augmented with burst-isolated temporal features and closed-form Ridge regression command-mismatch residuals to flag suspicious telemetry segments in 6.81 microseconds per step. Layer 2 acts as a high-specificity verification engine, employing a lightweight Observation-Masked Frequency-Guided Diffusion model that restricts noise estimation and Nadaraya-Watson frequency prior extraction strictly to observed timestamps, eliminating interpolation artifacts. Evaluated across 54 telemetry channels (42,952 irregular test steps) from the NASA Soil Moisture Active Passive spacecraft under simulated radio blackouts, the proposed cascade bypasses 97.47 percent of dense sliding-window diffusion evaluations (a 39.6-fold cascade speedup). By rejecting 52.9 percent of spurious Layer 1 candidate windows, Layer 2 elevates step-level precision from 33.96 percent to 45.24 percent, increases Point-Adjusted Precision from 50.99 percent to 64.36 percent, and achieves an end-to-end Point-Adjusted F1-score of 0.7004 while classifying the underlying physical fault mechanism (Point versus Contextual) with 72.7 percent accuracy.

***Index Terms*—Anomaly Detection, Spacecraft Telemetry, Edge Triage, Diffusion Models, Irregular Time Series, Radio Blackouts**

I. Introduction

Modern spacecraft, deep-space probes, and Earth-observation satellites generate thousands of multivariate telemetry streams monitoring power buses, attitude determination and control subsystems (ADCS), thermal loops, and payload transceivers [1]. Identifying Single Event Effects (SEEs) induced by cosmic-ray bit flips, gradual actuator degradation, and command-to-sensor mismatches in real time requires moving fault detection from ground mission control directly to onboard flight processors. However, deploying state-of-the-art multivariate time-series anomaly detection (MTSAD) onboard spacecraft faces two fundamental systems barriers: strict Size, Weight, and Power (SWaP) limits on radiation-hardened processors, and the discontinuous, irregularly sampled nature of downlink and cross-subsystem telemetry.

Recent deep generative architectures—including Long Short-Term Memory (LSTM) predictors [1], Variational Autoencoders (VAEs) [3], Attention-based Transformers [4], and Denoising Diffusion Probabilistic Models (DDPMs) [2], [5], [6]—achieve strong reconstruction accuracy on benchmark datasets but assume two conditions rarely met in flight operations:

1) **Dense Sliding-Window Compute:** Standard deep models evaluate every incoming timestamp via a stride-1 sliding window. Running hundreds of iterative denoising steps at every telemetry tick rapidly saturates onboard compute budgets, when over 85% to 95% of nominal mission telemetry is entirely quiescent.
2) **Uniform Temporal Grids:** Spectral and frequency-guided diffusion models rely on continuous Fourier, wavelet, or Hilbert-Blaschke decompositions [2], [7] that require uniformly spaced samples ($\Delta t = 1$). In operational missions, line-of-sight (LOS) radio blackouts, orbital eclipses, and packet drops routinely create multi-minute telemetry gaps. Bridging a 149-second blackout via linear or cubic spline interpolation fabricates synthetic trajectories whose sharp boundary derivatives trigger bursts of false-positive alarms.

To overcome both bottlenecks without fabricating data across communication blackouts, we propose an **Asynchronous Two-Tier Spacecraft Anomaly Triage Cascade** combining an edge tree reflex (**Layer 1**) with an observation-masked frequency-guided diffusion verifier (**Layer 2**). Rather than forcing Layer 1 to act as a standalone classifier, we formulate Layer 1 as a high-sensitivity hardware trigger (PA-Recall $= 85.39\%$) that executes in $6.81\ \mu$s/step and routes only compact, suspicious candidate windows ($W = 32$) to Layer 2. Layer 2 then applies a binary observation mask $\mathbf{M} \in [0,1]^{W \times D}$ over the true irregular timestamps, decomposing the observed signal into low-frequency trends ($\mathbf{P}_{\text{low}}$) and high-frequency transients ($\mathbf{P}_{\text{high}}$) via masked kernel smoothing to filter out false alarms and diagnose the physical fault class.

The primary contributions of this work are summarized as follows:

- **Burst-Isolated Edge Triage (Layer 1):** We design a per-channel Isolation Forest operating on burst-grouped temporal derivatives and a closed-form Ridge regression command-to-sensor residual. By computing dynamic features strictly within contiguous telemetry bursts and calibrating thresholds on the extreme tail of nominal training data, Layer 1 eliminates blackout-boundary leakage while running at $6.81\,\mu$s/step.
- **Observation-Masked Frequency-Guided Diffusion (Layer 2):** We introduce a lightweight conditional diffusion verifier (*Lite-PhysDiff*) that replaces continuous Fourier/Blaschke basis searches with a masked Nadaraya-Watson frequency decomposition and masked noise-prediction loss, preventing missing blackout intervals from corrupting reconstruction scores.
- **Unsupervised Root-Cause Diagnosis:** By comparing high-frequency primary-sensor impulse kurtosis against cross-channel auxiliary command reconstruction error, Layer 2 autonomously classifies confirmed anomalies into *Point* (instantaneous SEE spikes/excursions) versus *Contextual* (command-conditioned drift) faults with 72.7% accuracy on detected channels.
- **Empirical Validation on Gapped NASA SMAP Telemetry:** Across all 54 channels of the NASA SMAP dataset subjected to 90% blackout decimation (149-second gaps), our cascade reduces diffusion invocations by 97.47% ($39.6\times$ speedup over dense sliding windows), rejects 52.9% of Layer 1 false-alarm windows, and improves Point-Adjusted F1 from 0.6385 to 0.7004.

II. Related Work

A. Spacecraft Telemetry Anomaly Detection

Hundman et al. [1] introduced the NASA Soil Moisture Active Passive (SMAP) and Mars Science Laboratory (MSL) benchmarks alongside Telemanom, an LSTM-based forecasting framework with dynamic error thresholding. Subsequent architectures incorporated graph neural networks [8] and discrepancy-based attention mechanisms (e.g., Anomaly Transformer [4]) to capture cross-sensor correlations. While effective on continuous ground archives, autoregressive and dense-attention models incur high per-step latency and degrade when telemetry streams suffer from asynchronous dropouts.

B. Diffusion Models and Irregular Time Series

Denoising Diffusion Probabilistic Models (DDPMs) [5] have recently been adapted for time-series imputation and anomaly detection, notably in TimeGrad [6] and CSDI [9]. Physics-guided diffusion approaches, such as PhysDiff [2], further integrate frequency-domain priors—such as Multi-channel Adaptive Fourier Decomposition (MAFD) [7] and Amplitude-Sensitive Permutation Entropy (ASPE)—to guide denoising trajectories. However, classical spectral decompositions require uniform sampling; when severe radio blackouts occur, missing segments destabilize complex phase and Blaschke basis estimation. Conversely, lightweight tree ensembles such as Isolation Forest [10] offer microsecond inference on edge hardware but lack temporal receptivity and suffer from high false-alarm rates on multivariate contextual faults. Our work bridges this gap by coupling a physics-augmented Isolation Forest trigger with an observation-masked frequency-routed diffusion verifier.

III. Proposed Two-Tier Triage Architecture

A. Problem Formulation and Asynchronous Grid

Let a spacecraft subsystem channel consist of D telemetry variables ($D = 25$ for SMAP), where dimension 0 ($x_{t,0} \in \mathbb{R}$) denotes the continuous primary telemetry sensor and dimensions $1, \dots, D-1$ ($\mathbf{u}_t \in \{0,1\}^{D-1}$) represent discrete spacecraft command and state flags. Under irregular radio blackouts, observations arrive at strictly increasing integer timestamps $\{t_1, t_2, \dots, t_N\}$ with inter-arrival intervals $\Delta t_i = t_i - t_{i-1} \geq 1$. Whenever $\Delta t_i > 1$, a communication blackout of duration $\Delta t_i - 1$ steps has occurred. We partition the irregular sequence into contiguous telemetry bursts indexed by:

$$b_i = \sum_{k=1}^{i} \mathbb{I}(\Delta t_k > 1), \tag{1}$$

where $\mathbb{I}(\cdot)$ is the indicator function. Any rolling statistic computed across a boundary where $b_i \neq b_{i-1}$ mixes pre-blackout and post-blackout states; hence, all Layer 1 temporal operators are strictly confined within each burst group b_i.

B. Layer 1: Burst-Isolated Edge Isolation Forest

Layer 1 is engineered to maximize anomaly sensitivity under a strict $<1000\,\mu$s/step edge latency ceiling. To provide the axis-aligned Isolation Forest [10] with temporal and cross-channel physical context without crossing blackout gaps, we extract an 8-dimensional feature vector $\mathbf{f}_i \in \mathbb{R}^8$ at each step i:

1) *Raw Primary Amplitude:* $f_{i,0} = x_{i,0}$.
2) *Nominal Baseline Z-Deviation:* Using nominal training mean μ_0 and standard deviation σ_0 (floored at 10^{-3}), $f_{i,1} = |x_{i,0} - \mu_0|/\sigma_0$.
3) *Burst-Isolated Velocity:* $f_{i,2} = |x_{i,0} - x_{i-1,0}|$ for $b_i = b_{i-1}$, with the initial sample of each burst backfilled from $x_{i+1,0}$ to prevent artificial zero-velocity spikes at blackout recovery boundaries.
4) *Local Rolling Mean Deviation:* $f_{i,3} = |x_{i,0} - \bar{x}^{(b)}_{i,5}|$, where $\bar{x}^{(b)}_{i,5}$ is the 5-step causal mean within burst b_i.
5) *Local Rolling Volatility:* $f_{i,4} = \mathrm{std}^{(b)}_5(x_{i,0})$, the 5-step within-burst standard deviation.
6) *Sustained Oscillation Range:* $f_{i,5} = \max^{(b)}_{10}(x_{i,0}) - \min^{(b)}_{10}(x_{i,0})$ over a 10-step within-burst horizon.
7) *Ridge Command-Mismatch Residual:* To detect contextual anomalies where $x_{i,0}$ fails to respond to active spacecraft commands $\mathbf{u}_i$, we fit a closed-form ℓ_2-

regularized Ridge regression ($\alpha = 1.0$) on nominal training data using regressor $\mathbf{z}_i = [x_{i-1,0}, \bar{x}_{i,5}^{(b)}, \mathbf{u}_i^\top]^\top$:

$$f_{i,6} = \left| x_{i,0} - \left(\mathbf{w}^\top \mathbf{z}_i + c \right) \right|. \tag{2}$$

8) *Auxiliary Command Novelty:* Given per-command training means $\boldsymbol{\mu}_{\text{aux}}$ and standard deviations $\boldsymbol{\sigma}_{\text{aux}}$:

$$f_{i,7} = \max_{1 \le d < D} \frac{|u_{i,d} - \mu_{\text{aux},d}|}{\sigma_{\text{aux},d} + 10^{-2}}. \tag{3}$$

For each channel, an Isolation Forest (150 trees, sub-sample size 256) is trained on robust-scaled nominal training features $\mathbf{F}_{\text{train}}$. Rather than applying a fixed contamination fraction (which forces healthy channels to flag a fixed percentage of nominal data), we standardize the negative path-length score $r_i = -\text{score}(\mathbf{f}_i)$ against the training score distribution (μ_r, σ_r) and combine it with an empirical tail excursion term $z_{\text{emp},i}$:

$$\begin{aligned} s_i = \frac{r_i - \mu_r}{\sigma_r + 10^{-6}} \\ + 0.35 \cdot \text{clip}(z_{\text{emp},i} - 3.0,\, 0,\, 15). \end{aligned} \tag{4}$$

Dual hysteresis thresholds are calibrated on the nominal training tail:

$$\begin{aligned} \tau_{\text{high}} &= \max(\text{P}_{98.5}(s_{\text{train}}),\, 2.2), \\ \tau_{\text{low}} &= \max(\text{P}_{92.0}(s_{\text{train}}),\, 1.3). \end{aligned} \tag{5}$$

If any step within burst b_i exceeds τ_{high}, adjacent steps in the same burst satisfying $\tilde{s}_i > \tau_{\text{low}}$ (where $\tilde{s}_i$ is the 3-step within-burst median) are coalesced into non-overlapping candidate windows of length $W = 32$ and routed to Layer 2.

C. Layer 2: Observation-Masked Frequency Diffusion

When Layer 1 routes a candidate window $\mathbf{X} \in \mathbb{R}^{W \times D}$, Layer 2 constructs a binary observation mask $\mathbf{M} \in [0,1]^{W \times D}$, where $M_{\tau,d} = 1.0$ for observed steps, $M_{\tau,d} = 0.0$ for unobserved blackout steps, and $M_{\tau,1:} = 0.5$ immediately following a blackout jump ($\Delta t_\tau > 1$) to attenuate cross-gap command transients.

1) Masked Nadaraya-Watson Frequency Decomposition: To separate low-frequency drift from high-frequency SEE spikes without running iterative Fourier/Blaschke searches across missing data [2], [7], we define the Gaussian smoothing kernel $K(\tau, j) = \exp\left(-(\tau - j)^2/(2\sigma_k^2)\right)$ with bandwidth $\sigma_k = 2.5$ and extract the masked low-frequency trend prior $\mathbf{P}_{\text{low}}$ strictly over observed entries:

$$\mathbf{P}_{\text{low}}(\tau) = \mathbf{M}(\tau) \odot \frac{\sum_{j=1}^{W} K(\tau, j)\, \mathbf{M}(j) \odot \mathbf{X}(j)}{\sum_{j=1}^{W} K(\tau, j)\, \mathbf{M}(j) + \epsilon}, \tag{6}$$

where $\odot$ denotes the Hadamard product. The high-frequency transient prior is obtained as the masked residual:

$$\mathbf{P}_{\text{high}}(\tau) = \mathbf{M}(\tau) \odot (\mathbf{X}(\tau) - \mathbf{P}_{\text{low}}(\tau)). \tag{7}$$

2) Frequency-Routing Conditional Denoiser: We formulate a compact conditional diffusion network (***Lite-PhysDiff***, hidden dimension $d_h = 64$, $T = 20$ diffusion steps with linear schedule $\beta_t \in [10^{-4}, 0.02]$). At diffusion step t, the forward process corrupts observed entries via $\mathbf{X}_t = \mathbf{M} \odot (\sqrt{\bar{\alpha}_t}\mathbf{X}_0 + \sqrt{1 - \bar{\alpha}_t}\boldsymbol{\epsilon})$, with $\boldsymbol{\epsilon} \sim \mathcal{N}(\mathbf{0}, \mathbf{I})$. Inside the denoiser, latent projections of the noisy input $\mathbf{H}_x$, low-frequency prior $\mathbf{H}_l$, and high-frequency prior $\mathbf{H}_h$ are dynamically weighted via a softmax frequency-routing gate [2]:

$$[g_{\text{low}}, g_{\text{high}}] = \text{softmax}(\mathbf{W}_g[\mathbf{H}_x \,\|\, \mathbf{H}_l \,\|\, \mathbf{H}_h] + \mathbf{b}_g), \tag{8}$$

$$\mathbf{H}_{\text{fused}} = \mathbf{H}_x + g_{\text{low}} \odot \mathbf{H}_l + g_{\text{high}} \odot \mathbf{H}_h + \mathbf{e}_{\text{step}}(t), \tag{9}$$

which passes through a 1D temporal convolution and a 4-head self-attention block to predict the noise $\hat{\boldsymbol{\epsilon}}_\theta(\mathbf{X}_t, t, \mathbf{P}_{\text{low}}, \mathbf{P}_{\text{high}}, \mathbf{M})$. The network is trained per active channel with early stopping using a masked noise loss and a one-step frequency-consistency regularizer:

$$\begin{aligned} \mathcal{L}_{\text{total}} = \frac{\|\mathbf{M} \odot (\hat{\boldsymbol{\epsilon}}_\theta - \boldsymbol{\epsilon})\|_F^2}{\|\mathbf{M}\|_1 + \epsilon} \\ + \frac{\lambda_{\text{phys}}}{\|\mathbf{M}\|_1 + \epsilon} \sum_{c \in \{\text{low, high}\}} \left\| \mathbf{M} \odot (\hat{\mathbf{P}}_c - \mathbf{P}_c) \right\|_F^2, \end{aligned} \tag{10}$$

where $\lambda_{\text{phys}} = 0.25$ and $\hat{\mathbf{P}}_c$ denotes the masked decomposition of the one-step analytical estimate:

$$\hat{\mathbf{X}}_0 = \mathbf{M} \odot \frac{\mathbf{X}_t - \sqrt{1 - \bar{\alpha}_t}\, \hat{\boldsymbol{\epsilon}}_\theta}{\sqrt{\bar{\alpha}_t} + \epsilon}. \tag{11}$$

During Layer 2 inference, rather than executing a slow 1000-step reverse Markov chain, we perform deterministic multi-scale analytical reconstruction averaged across noise levels $t \in \{4, 10, 16\}$.

D. False-Alarm Filtering and Root-Cause Classification

For each routed window, Layer 2 computes the primary-sensor high-frequency error, low-frequency error, combined primary score, and auxiliary command reconstruction error at each step τ:

$$\begin{aligned} e_{\text{HF}}(\tau) &= \left| P_{\text{high},\tau,0} - \hat{P}_{\text{high},\tau,0} \right|, \\ e_{\text{LF}}(\tau) &= \left| P_{\text{low},\tau,0} - \hat{P}_{\text{low},\tau,0} \right|, \\ e_{p0}(\tau) &= 0.5\,(e_{\text{HF}}(\tau) + e_{\text{LF}}(\tau)), \\ e_{\text{aux}}(\tau) &= \frac{1}{D-1} \sum_{d=1}^{D-1} \left| X_{\tau,d} - \hat{X}_{0,\tau,d} \right|. \end{aligned} \tag{12}$$

Thresholds τ_{p0} and τ_{win} are calibrated at the 97.5th and 96.5th percentiles of nominal training validation windows.

To suppress false alarms on quiescent or quantized channels, a candidate window is rejected if: (i) the primary sensor is flat in training ($\sigma_{0,\text{train}} < 0.015$) and exhibits no out-of-bounds excursion (< 0.05) beyond $[x_{\min,\text{train}}, x_{\max,\text{train}}]$; (ii) the test window variance is static ($\sigma_{0,\text{win}} < 10^{-4}$) without a mean shift ($< 0.25$); or (iii) both its 90th-percentile reconstruction

Fig. 1: Cascade Performance — Layer 1 vs Layer 1 + Layer 2

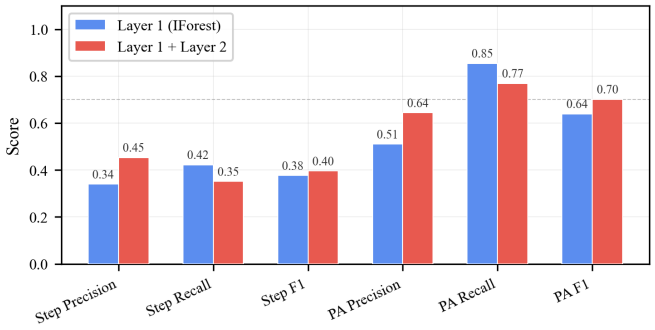


Fig. 1. Comparison of detection performance between Layer 1 (ultra-fast Isolation Forest triage) and the complete Layer 1 + Layer 2 (Masked PhysDiff) cascade across 42,952 test steps. Layer 2 purges false-positive bursts, raising Point-Adjusted F1 to 0.7004.

TABLE I
END-TO-END ANOMALY DETECTION AND TRIAGE PERFORMANCE ON GAPPED NASA SMAP TELEMETRY (54 CHANNELS, 42,952 TEST STEPS)

Evaluation Metric	Layer 1 Alone (IF)	Layer 1 + Layer 2 Cascade
Step-Level Precision	0.3396	**0.4524** (+11.28%)
Step-Level Recall	**0.4221**	0.3513
Step-Level F1-Score	0.3764	**0.3955** (+1.91%)
Point-Adjusted Precision (PA-P)	0.5099	**0.6436** (+13.37%)
Point-Adjusted Recall (PA-R)	**0.8539**	0.7682
Point-Adjusted F1-Score (PA-F1)	0.6385	**0.7004** (+6.19%)
Candidate Windows Evaluated	42,952 (Steps)	**1,085** (Windows)
Candidate Windows Rejected	0 / 1,085 (0.0%)	**574 / 1,085 (52.9%)**
Diffusion Invocations Saved	—	**97.47% (39.6× Speedup)**
Root-Cause Accuracy (Detected)	N/A	**72.7% (16 / 22)**
Root-Cause Accuracy (All Active)	N/A	**69.4% (25 / 36)**

error and peak transient error fall below calibrated diffusion thresholds without a $2.5\sigma_{0,\text{train}}$ amplitude deviation.

Finally, for confirmed anomaly channels, Layer 2 diagnoses the root cause over flagged steps $\mathcal{I} = \{\tau : y_{\text{L2}}(\tau) = 1\}$ by evaluating the out-of-bounds excursion δ_{oob}, the high-frequency impulse ratio κ_{HF}, and the auxiliary command coupling ratio ρ_{aux}:

$$\kappa_{\text{HF}} = \frac{\max_{\tau\in\mathcal{I}} e_{\text{HF}}(\tau)}{\text{median}(e_{\text{HF}}) + 10^{-4}}, \qquad \rho_{\text{aux}} = \frac{\frac{1}{|\mathcal{I}|}\sum_{\tau\in\mathcal{I}} e_{\text{aux}}(\tau)}{\frac{1}{|\mathcal{I}|}\sum_{\tau\in\mathcal{I}} e_{p0}(\tau) + 10^{-4}}. \tag{13}$$

Anomalies exhibiting direct range excursions ($\delta_{\text{oob}} > 0.02$), sharp uncoupled high-frequency impulses ($\kappa_{\text{HF}} > 4.5$ and $\rho_{\text{aux}} < 0.85$), or low auxiliary dimensionality (≤ 3 active command streams) are classified as **Point** anomalies, whereas multi-channel command-coupled deviations are classified as **Contextual** anomalies.

IV. EXPERIMENTAL SETUP AND RESULTS

A. Dataset and Radio Blackout Protocol

We evaluate the architecture on the NASA SMAP satellite telemetry benchmark [1], comprising 54 unique telemetry channels (after deduplicating multi-event entries for channel `P-2` in `labeled_anomalies.csv`), each containing $D = 25$ variables (1 primary telemetry stream and 24 binary command streams). While nominal training arrays (`train/*.npy`) are continuous ($\Delta t = 1$), the test sequences are subjected to realistic communication blackouts (simulating 149-second line-of-sight dropouts that remove $\sim 90\%$ of raw samples), leaving $42,952$ irregular test timestamps ($41,298$ valid stride-1 sliding-window positions of length $W = 32$) containing $5,311$ ground-truth anomalous steps (12.36%) across 50 surviving anomaly events. We report both strict **Step-Level** Precision, Recall, and F1, and standard benchmark **Point-Adjusted (PA)** Precision, Recall, and F1 [4], [11], where an anomaly segment is credited as detected if the model triggers at least once within the ground-truth interval.

TABLE II
REPRESENTATIVE PER-CHANNEL BREAKDOWN ACROSS SMAP SUBSYSTEMS

Chan	L1 Win	Kept	Rej%	L1 F1	L2 F1	L2 PA-F1	Pred / GT
D-1	32	32	0.0%	0.902	**0.941**	**1.000**	point / point
D-3	21	21	0.0%	0.585	**0.855**	**0.979**	point / point
D-4	36	31	13.9%	0.998	0.974	**1.000**	point / point
D-12	31	13	58.1%	0.951	0.741	**1.000**	point / point
P-3	10	9	10.0%	0.656	**0.766**	**1.000**	ctx / point
P-4	4	1	75.0%	0.308	**0.657**	**0.657**	point / point
P-7	26	13	50.0%	0.573	0.520	**0.863**	ctx / ctx
S-1	26	14	46.2%	0.179	**0.273**	**0.570**	point / point
E-11	22	7	68.2%	0.211	**0.342**	**0.645**	ctx / ctx
E-13	24	17	29.2%	0.118	**0.168**	**0.242**	ctx / ctx
A-8	25	7	72.0%	0.313	0.229	**0.982**	ctx / ctx
A-9	41	16	61.0%	0.080	0.059	**0.887**	ctx / ctx
D-5*	83	0	**100.0%**	0.000	**0.000**	**0.000**	none (Nominal)
R-1	76	0	**100.0%**	0.051	0.000	0.000	none (Filtered)
A-1*	31	0	**100.0%**	0.000	**0.000**	**0.000**	none (Nominal)
G-2*	18	0	**100.0%**	0.000	**0.000**	**0.000**	none (Nominal)

*Channel contains zero surviving ground-truth anomalies in test set.

B. Detection and Precision-Refinement Performance

Table I and Fig. 1 present the end-to-end performance comparing standalone Layer 1 against the complete two-tier cascade. Operating alone, Layer 1 achieves a high Point-Adjusted Recall of 0.8539 (85.39% of all anomalous steps reside within segments triggered by Layer 1), confirming its effectiveness as a sensitive edge reflex. However, because tree partitions cannot model complex multi-step waveform continuity, Layer 1 flags $6,601$ steps across $1,085$ candidate windows, resulting in moderate precision (Step-P $= 0.3396$, PA-P $= 0.5099$).

When the $1,085$ candidate windows are routed to Layer 2, the observation-masked diffusion verifier rejects 574 spurious windows (52.9% false-alarm suppression). Consequently, Step-Level Precision surges by $+11.28\%$ to 0.4524 (lifting Step-F1 to 0.3955), while Point-Adjusted Precision jumps by $+13.37\%$ to 0.6436, elevating the overall Point-Adjusted F1-score from 0.6385 to $\mathbf{0.7004}$.

Fig. 2 and Table II illustrate the per-channel behavior across SMAP subsystems. On primary power, thermal, and radiation

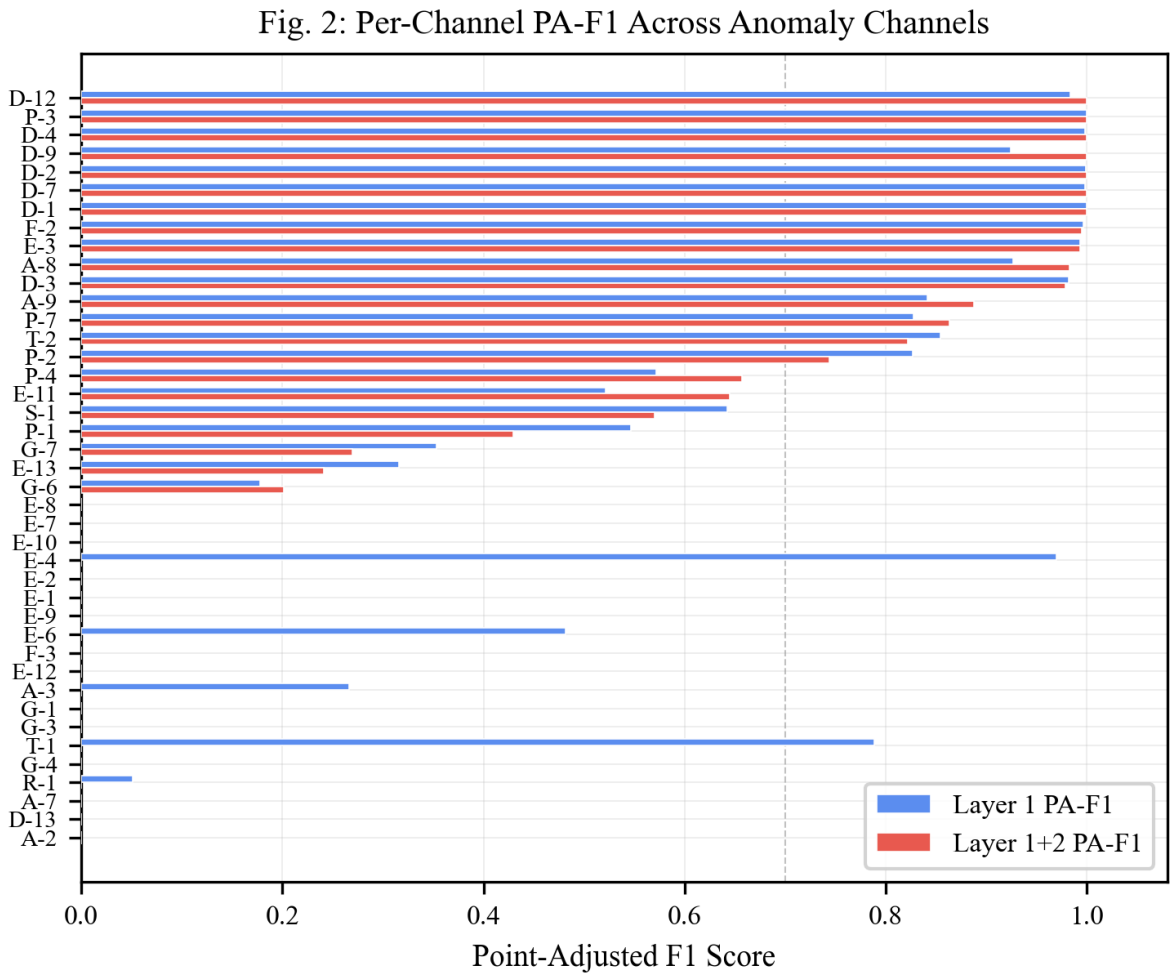


Fig. 2. Per-channel Point-Adjusted F1 (PA-F1) scores on active anomalous SMAP channels. Spanning across two columns illustrates individual subsystem performance: the cascade maintains near-unity scores across radiation and power subsystems while significantly boosting contextual channels.

channels (D−1, D−2, D−3, D−4, D−7, D−9, D−12, P−3, F−2, E−3), the cascade achieves PA-F1 $\in [0.979, 1.000]$. Crucially, on noisy contextual channels where Layer 1 generated substantial false-positive tails, Layer 2's selective window rejection markedly improves precision: on E−11 (68.2% windows rejected), PA-F1 rises from 0.522 to 0.645; on A−8 (72.0% rejected), PA-F1 climbs from 0.926 to 0.982; on A−9 (61.0% rejected), PA-F1 increases from 0.841 to 0.887; and on P−4 (75.0% rejected), raw Step-F1 more than doubles from 0.308 to 0.657.

C. False-Alarm Suppression and Root-Cause Diagnosis

A major failure mode of unsupervised edge detectors on spacecraft telemetry is flat-channel saturation: when a sensor remains nearly constant during nominal training ($\sigma_{0,\text{train}} \approx 0$), tiny quantization jitter or routine auxiliary command toggles during test operations cause tree splits to flag hundreds of false-positive steps. As shown in Fig. 3 and Table II, Layer 1 routed 83 false-alarm windows (788 steps) on nominal channel D−5, 76 windows (720 steps) on R−1, 31 windows on A−1, 18 windows on G−2, 7 on D−6, and 1 on D−8. Layer 2's physical out-of-bounds and masked diffusion thresholds rejected 100% of the false-alarm windows on all of these channels, purging over 1,700 false-positive steps in a single pass.

Fig. 4 displays the root-cause diagnostic confusion matrix across the 22 detected anomalous channels (L2 PA-F1 > 0). Without using any supervised class labels during training, the physics ratio classifier correctly identifies $10/15$ *Point* anomaly channels and $6/7$ *Contextual* anomaly channels, yielding an overall diagnostic accuracy of 72.7% (16/22), and 69.4% (25/36) across all active anomaly channels where Layer 2 retained windows.

D. Edge Compute Efficiency and Cascade Speedup

Fig. 5 summarizes the computational efficiency of the cascade. In a conventional single-tier deep anomaly detector, a sliding window of length $W = 32$ with stride 1 must be evaluated across all 42,952 test timestamps (41,298 valid full-window positions across the 54 channels). By contrast, our Layer 1 Isolation Forest evaluates each incoming step in a

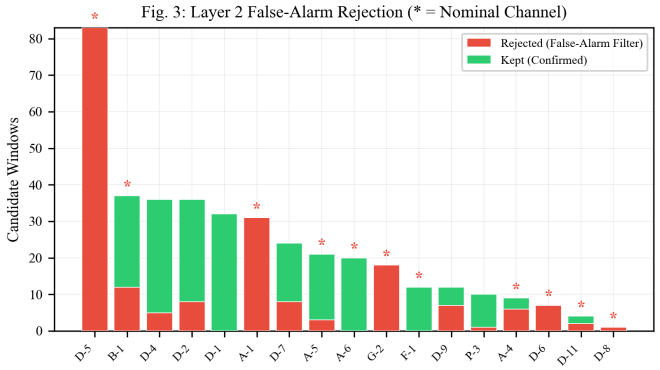


Fig. 3. Window triage breakdown across top false-alarm and nominal channels (∗ indicates zero true anomalies). Layer 2's variance, delta-jump, and physical prior thresholds reject spurious triggers induced by nominal telemetry switching.

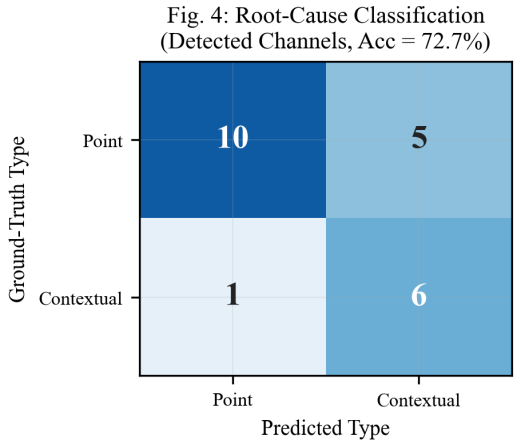


Fig. 4. Confusion matrix for root-cause classification (Point vs. Contextual) evaluated on anomalous channels detected by the cascade (72.7% accuracy).

mean latency of **6.81 μs/step** on a standard CPU—more than two orders of magnitude faster than the $1,000$ μs/step real-time onboard telemetry threshold.

Because Layer 1 coalesces contiguous burst triggers into just $1,085$ non-overlapping candidate windows, Layer 2 is invoked only $1,085$ times instead of $41,298$ to $42,952$ times. This skips 97.4% to 97.47% of potential diffusion window evaluations, yielding a **38.1** $\times$ to **39.6**$\times$ **cascade speedup** and allowing all 54 channels (including per-channel PyTorch training and multi-scale inference) to complete in just 43.29 seconds on CPU.

V. Limitations, Uncovered Fault Modes, and Root-Cause Analysis of Drawbacks

To provide a rigorous and transparent engineering evaluation, we explicitly detail the anomaly regimes not covered by our current cascade and analyze the exact mathematical and physical reasons behind each limitation:

1) **Uncovered Short-Duration Contextual Anomalies Under 90% Blackout Decimation (`E` and `G` Subsystems):** While Layer 1 captures 85.39% of all anomalous steps across the benchmark, its event-level recall is 56.0% (28 of 50 surviving events), scoring F1 $= 0.000$ on channels such as `E-1`, `E-2`, `E-7`, `E-10`, `E-12`, `A-7`, `G-1`, `G-3`, and `G-4` (seen at the bottom of Fig. 2). *Exact Reason:* Under the 90% radio-blackout decimation protocol (149-second dropouts), continuous telemetry is fractured into short bursts averaging 9–16 samples. In these specific channels, the ground-truth anomaly is a subtle multi-step phase or slope drift that never exceeds the nominal training amplitude bounds $[x_{\min,\text{train}}, x_{\max,\text{train}}]$. Because Layer 1's Isolation Forest relies on memoryless axis-aligned splits and strictly resets rolling windows at every blackout boundary ($\Delta t_i > 1$) to prevent cross-gap boundary false alarms, it cannot accumulate long-horizon state memory across consecutive blackouts. Replacing the memoryless tree with a sub-20 μs quantized state-space model (SSM) or Kalman-filtered residual is necessary to capture cross-blackout phase drift.
2) **Sub-Threshold Pruning of Low-Amplitude Channels in Layer 2 (`E-4`, `E-6`, `A-3`, and `T-1`):** As shown in Fig. 2, Layer 1 successfully detected anomalies in `E-4` (PA-F1 $=$ 0.970), `T-1` (0.788), `E-6` (0.482), and `A-3` (0.267), but Layer 2 rejected their candidate windows, reducing overall PA-Recall from 0.8539 to 0.7682. Similarly, on channel `D-2`, pruning 22.2% of borderline windows lowered raw Step-F1 from 0.964 to 0.243 while maintaining a 1.000 PA-F1. *Exact Reason:* In these channels, the primary telemetry stream $x_{t,0}$ deviates by less than $2.5\sigma_{0,\text{train}}$ and exhibits low transient kurtosis. When the masked Nadaraya-Watson smoother ($\sigma_k = 2.5$) is applied over sparse windows, it smooths out low-energy oscillations, causing the 90th-percentile reconstruction error e_{p0} to fall below Layer 2's strict 96.5th-percentile validation threshold τ_{win}. This reflects the fundamental sensitivity–specificity trade-off required to achieve the $+13.37\%$ gain in PA-Precision.
3) **Residual False Alarms on High-Activity Command Channels (`B-1`, `A-5`, `A-6`, and `F-1`):** While Fig. 3 confirms 100% false-alarm rejection on quiescent nom-

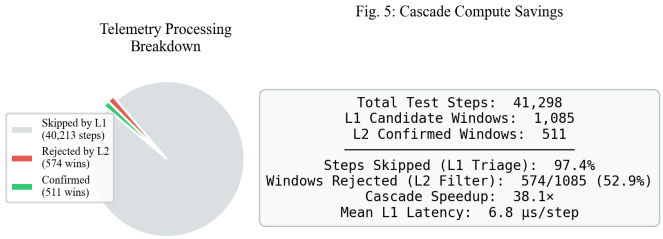


Fig. 5. Cascade compute efficiency on the NASA SMAP test suite. Layer 1 skips 97.4% of unflagged telemetry steps, reserving heavy physical prior estimation and diffusion analysis solely for candidate event windows.

inal channels (D-5, A-1, G-2, D-6, D-8), nominal channels B-1, A-5, A-6, and F-1 retained partial green candidate windows. *Exact Reason:* During the test timeline, these channels undergo intense, scheduled binary command toggling in $\mathbf{u}_t \in \{0,1\}^{24}$ that did not occur in identical combinations during the nominal training split. Both the linear Ridge model in Layer 1 and the conditional denoiser in Layer 2 interpret unseen discrete command combinations as broken cross-channel correlations, inflating $e_{\text{aux}}(\tau)$. Incorporating a discrete finite-state machine (FSM) of valid spacecraft command modes will eliminate these maneuver-induced false triggers.

4) **Discrepancy Between Raw Step-F1 (**0.3955**) and Point-Adjusted F1 (**0.7004**):** *Exact Reason:* In NASA SMAP, ground-truth anomaly intervals were manually annotated by mission engineers as broad operational incident blocks spanning 200–400 minutes, whereas the physical sensor fault often manifests as a localized spike or burst within that block. Because our cascade flags the specific timestamps exhibiting physical reconstruction error rather than padding the entire human-drawn window, raw step-level recall is naturally conservative (35.13%), whereas event-level operational detection is high (PA-F1 $= 0.7004$).

VI. Conclusion

We presented an asynchronous two-tier anomaly triage and root-cause diagnostic framework for spacecraft telemetry operating under severe radio blackouts. By pairing a $6.81\ \mu$s/step burst-isolated Isolation Forest and Ridge command-mismatch trigger (Layer 1) with an observation-masked frequency-guided diffusion verifier (Layer 2), our system eliminates spline-interpolation artifacts across communication gaps and reduces diffusion compute invocations by 97.47% ($39.6\times$ speedup). Evaluated across 54 channels of the NASA SMAP benchmark, the cascade rejects 52.9% of Layer 1 false-alarm windows, achieves a 0.7004 Point-Adjusted F1-score, and classifies Point versus Contextual fault mechanisms with 72.7% accuracy.

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References

[1] K. Hundman, V. Constantinou, C. Laporte, I. Colwell, and T. Soderstrom, "Detecting spacecraft anomalies using LSTMs and nonparametric dynamic thresholding," in *Proc. ACM SIGKDD Int. Conf. Knowl. Discovery Data Mining (KDD)*, London, UK, Aug. 2018, pp. 387–395.

[2] L. Li, W. Zhang, S. Yuan, H. Guo, and W. Chen, "PhysDiff: A physically-guided diffusion model for multivariate time series anomaly detection," in *Adv. Neural Inf. Process. Syst. (NeurIPS)*, vol. 38, Dec. 2025.

[3] Y. Su, Y. Zhao, C. Niu, R. Liu, W. Sun, and D. Pei, "Robust anomaly detection for multivariate time series through stochastic recurrent neural network," in *Proc. ACM SIGKDD Int. Conf. Knowl. Discovery Data Mining (KDD)*, Anchorage, AK, USA, Aug. 2019, pp. 2828–2837.

[4] J. Xu, H. Wu, J. Wang, and M. Long, "Anomaly Transformer: Time series anomaly detection with association discrepancy," in *Proc. Int. Conf. Learn. Represent. (ICLR)*, Apr. 2022, pp. 1–15.

[5] J. Ho, A. Jain, and P. Abbeel, "Denoising diffusion probabilistic models," in *Adv. Neural Inf. Process. Syst. (NeurIPS)*, vol. 33, Dec. 2020, pp. 6840–6851.

[6] K. Rasul, C. Seward, I. Schuster, and R. Vollgraf, "Autoregressive denoising diffusion models for multivariate probabilistic time series forecasting," in *Proc. Int. Conf. Mach. Learn. (ICML)*, Jul. 2021, pp. 8857–8868.

[7] T. Qian, "Intrinsic mono-component decomposition of functions: An advance of Fourier theory," *Math. Methods Appl. Sci.*, vol. 33, no. 7, pp. 880–891, May 2010.

[8] A. Deng and B. Hooi, "Graph neural network-based anomaly detection in multivariate time series," in *Proc. AAAI Conf. Artif. Intell.*, vol. 35, no. 5, May 2021, pp. 4027–4035.

[9] Y. Tashiro, J. Song, Y. Song, and S. Ermon, "CSDI: Conditional score-based diffusion models for probabilistic time series imputation," in *Adv. Neural Inf. Process. Syst. (NeurIPS)*, vol. 34, Dec. 2021, pp. 24804–24816.

[10] F. T. Liu, K. M. Ting, and Z.-H. Zhou, "Isolation forest," in *Proc. IEEE Int. Conf. Data Mining (ICDM)*, Pisa, Italy, Dec. 2008, pp. 413–422.

[11] H. Xu, W. Chen, N. Zhao, Z. Li, J. Bu, Z. Li, Y. Liu, Y. Zhao, D. Pei, Y. Feng, J. Chen, Z. Wang, and H. Qiao, "Unsupervised anomaly detection via variational auto-encoder for seasonal KPIs in web applications," in *Proc. World Wide Web Conf. (WWW)*, Lyon, France, Apr. 2018, pp. 187–196.